Predicting SpaceX Falcon 9 First-Stage Landing Success

Expanded analysis report: EDA, SQL results, map and dashboard deliverables

Repository: https://github.com/sudharsan0317/spacex-capstone-project

Dataset: 101 launch records, dated 2010–2020. Target label is 1 when Landing_Outcome begins with “Success”; otherwise 0.

Scope note: The charts, SQL query results and baseline model metrics in this report were computed from the provided CSV. The Folium map and Dash application source are included as separate files in the project bundle. The map is an interactive HTML file; the dashboard source must be run locally with the listed Python packages. This does not claim that every original IBM course lab was completed.

1. Executive Summary

The goal is to explore whether historical launch attributes can help classify a successful first-stage landing label.

The dataset contains 101 records: 61 labelled successful and 40 not labelled successful under the stated binary rule.

Logistic Regression outperformed Random Forest on this one 75/25 stratified test split (accuracy 65.4% vs 61.5%). Results are preliminary because the dataset and test set are small.

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2. Data Preparation and Target

Predictors: payload mass, launch site, orbit, customer, and booster family. Numeric payload mass was median-imputed and standardized. Categorical variables were imputed using the most frequent value and one-hot encoded.

The landing label groups several different categories into non-success, including “No attempt” and various failures or controlled outcomes. This simplification is documented and should be considered when interpreting the model.

The test set contained 26 records. Because this is a small test set, a few changed predictions can noticeably change the reported metrics.

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3. Exploratory Data Analysis — Landing Outcomes

Landing outcome categories are not operationally identical. In particular, “No attempt” is distinct from a landing failure; keep this distinction in mind when reading the binary model results.

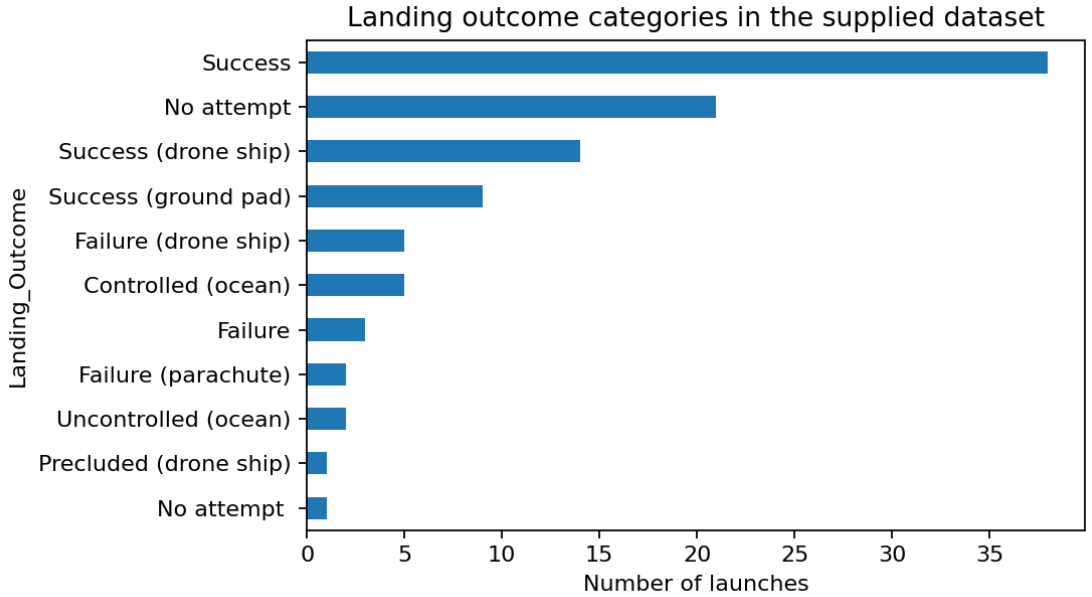


Figure 1. Counts by raw landing outcome category.

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4. Launch Site Analysis

The SQL results show the number of records per site and the share labelled successful. These percentages are descriptive only; site records may differ by era, booster, mission type and sample size.

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Launch site	Launches	Labelled successes	Labelled success %
KSC LC-39A	25	20	80.0%
CCAFS SLC-40	34	25	73.5%
VAFB SLC-4E	16	10	62.5%
CCAFS LC-40	26	6	23.1%

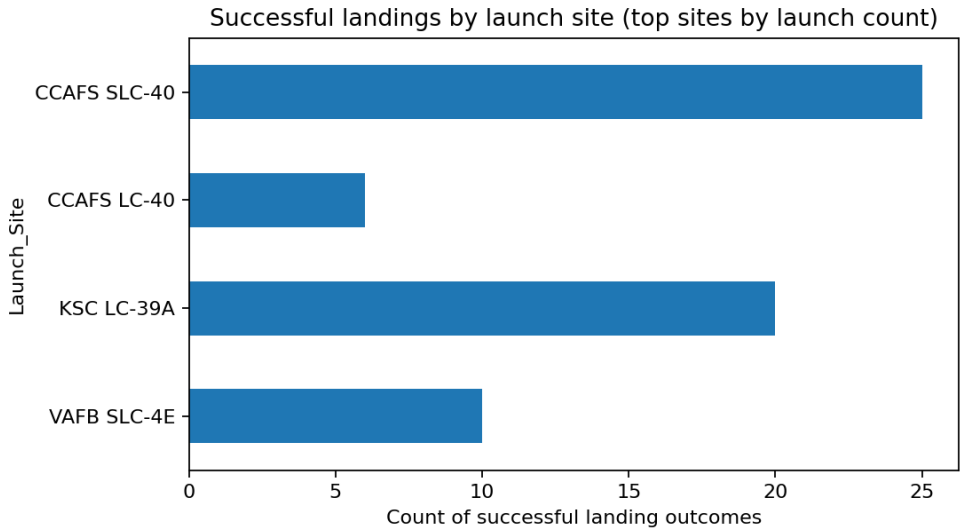


Figure 2. Successful-label counts by launch site. Counts are not rates.

5. Payload Mass Exploration

Payload mass is a potential predictor but the scatter plot cannot establish causation. The dataset includes zero-valued payload masses; confirm whether those are genuine values or placeholders before interpreting payload effects.

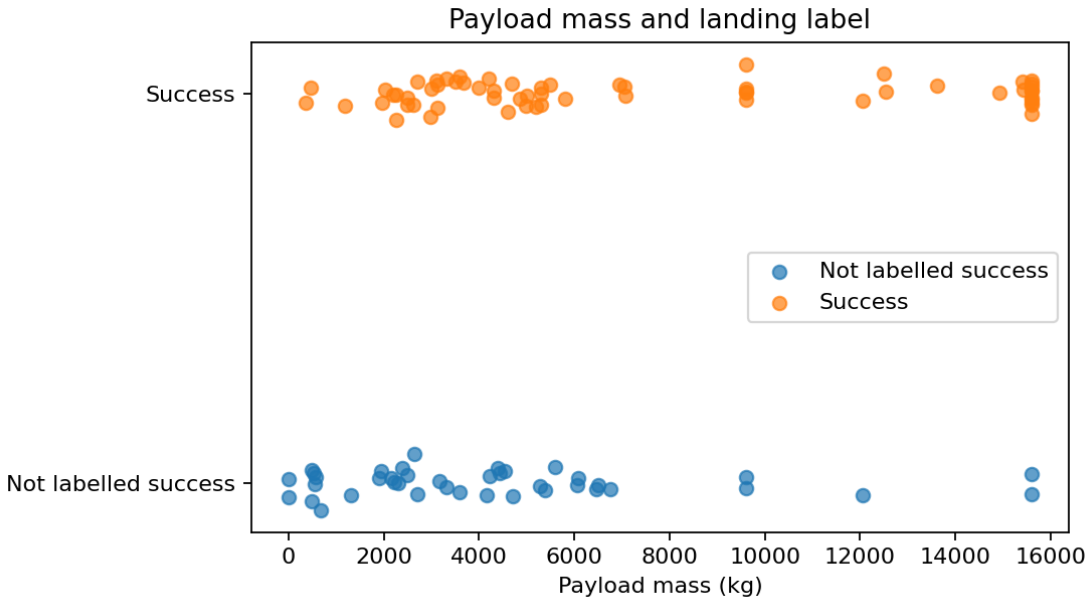


Figure 3. Payload mass by binary landing label.

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6. SQL Analysis — Queries and Findings

Query 1 — launches by site: grouped records by Launch_Site and counted launches. Results: CCAFS SLC-40 = 34; CCAFS LC-40 = 26; KSC LC-39A = 25; VAFB SLC-4E = 16.

Query 2 — labelled success by site: KSC LC-39A = 20/25 (80.0%); CCAFS SLC-40 = 25/34 (73.5%); VAFB SLC-4E = 10/16 (62.5%); CCAFS LC-40 = 6/26 (23.1%). These are dataset-derived descriptive percentages under this report's binary label rule.

Query 3 — outcome categories: counts by raw Landing_Outcome are saved in sql_outcome_counts.csv. Query text and full result tables are provided in sql_queries_and_results.txt and separate CSV files.

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7. Interactive Launch-Site Map

An interactive Folium map is included as spacex_launch_sites_map.html. Open it in a web browser to see markers for the four launch-site labels in this dataset. Click a marker for the record count and labelled-success percentage.

Coordinates are approximate site locations. CCAFS SLC-40 and CCAFS LC-40 are represented at the same general Cape Canaveral area; the map should not be interpreted as a precision engineering map.

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8. Dashboard Deliverable

The bundle includes spacex_dashboard.py, a Dash app with a launch-site dropdown, a summary of launch counts and labelled success, a landing-outcome histogram, and payload-mass plot.

To run it locally: install Python packages with `pip install dash plotly pandas`; place spacex_dashboard.py and Spacex.csv in the same folder; run `python spacex_dashboard.py`; open the localhost URL printed in the terminal. This source file is provided; it is not hosted as a live public dashboard.

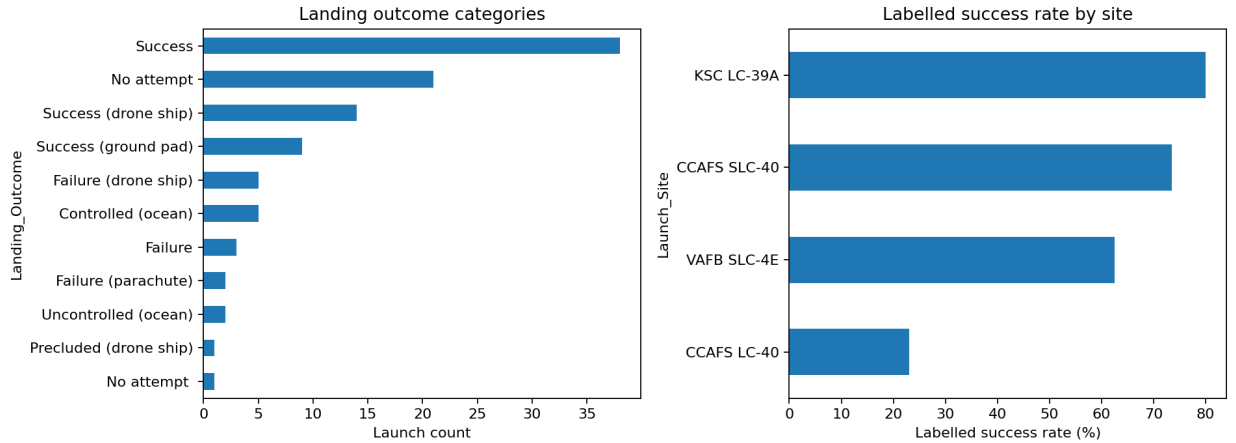


Figure 4. Static preview of dashboard-style analysis; the interactive app is supplied separately.

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9. Machine-Learning Results

Two baseline classifiers were evaluated on the same stratified train/test split. Metrics were computed on the held-out test set.

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Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.654	0.706	0.750	0.727
Random Forest	0.615	0.667	0.750	0.706



Figure 5. Logistic Regression confusion matrix on the test split.

10. Conclusions, Limitations and Next Steps

This project demonstrates data loading, cleaning, exploratory visualization, SQL aggregation, a geographic launch-site map, dashboard source, and a basic supervised classification workflow.

Limitations: only 101 records; a single train/test split; binary target simplification; no extensive hyperparameter tuning; no cross-validation; and no external validation. The model should not be used to make real launch decisions.

Recommended next steps: compare models with cross-validation; inspect class balance and zero payload values; add more data; explain model errors; and capture screenshots from running the interactive map and dashboard if the assignment explicitly requires evidence.

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